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Tiny Machine Learning (TinyML): Research trends and future application opportunities

Hui Han [a,*], Silvana Trimi [b], Sang M. Lee [b]

[a] Department of Computer Science, Electrical and Space Engineering, Luleå University of Technology, 97187, Luleå, Sweden
[b] Department of Supply Chain Management and Analytics, University of Nebraska, Lincoln, NE68588, USA

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ABSTRACT

Tiny Machine Learning (TinyML) enables artificial intelligence on low-power edge devices, yet a quantitative understanding of TinyML research remains limited. This study addresses this gap through a comprehensive bibliometric analysis of 392 peer-reviewed publications (2020–2024) from the Web of Science, using Biblioshiny and VOSviewer. This article contributes by mapping the first bibliometric structure of TinyML, identifying major trends (exponential publication growth, strong international collaboration, core research themes, key contributors, etc.) and proposing future directions (such as sustainable hardware, federated learning, ethical frameworks, etc.). The findings provide a scholarly foundation and strategic roadmap for advancing scalable, energy-efficient, and privacy-preserving TinyML applications.

1. Introduction

Tiny Machine Learning (TinyML) is an emerging field of artificial intelligence (AI) focused on deploying machine learning models on resource-constrained, low-powered devices. By integrating embedded systems and AI, TinyML facilitates local data processing on edge devices, reducing dependency on cloud computing while enhancing data privacy, decreasing latency, and minimizing power consumption [1]. Applications of TinyML span diverse fields of information systems, including environmental monitoring, healthcare, industrial automation, and smart city, where real-time data processing and privacy are critical.

Several review and survey papers on TinyML have been published in recent years, each emphasizing different aspects of this rapidly growing field. Lakshman and Eisty (2022) [2] focused on software engineering approaches for TinyML-based IoT vision systems, while Rajapakse et al. (2022) [3] examined model development workflows for microcontroller-class hardware. Schizas et al. (2022) [4] provided a systematic review of TinyML for large-scale IoT deployments, and Han and Siebert (2022) [5] synthesized early TinyML research across hardware, frameworks, datasets, and applications. More recently, Tsoukas et al. (2024) [6] offered a comprehensive technical survey emphasizing optimization techniques and hardware–software co-design. However, all prior studies remain primarily qualitative and lack quantitative insight into the global evolution, collaboration patterns, and thematic structure of TinyML research.

In contrast, this study presents the first bibliometric analysis of TinyML research, addresses this gap through the Web of Science (WoS) database–known for its extensive coverage of academic literature. A total of 471 articles were initially identified using the search string "tinyml" or "tiny machine learning" across various document types including journal articles, proceeding papers, and reviews. After applying inclusion criteria such as peer-reviewed sources, English language, and the publication time frame from January 2020 to May 2024, the dataset was refined to 392 articles. The final dataset was analyzed to identify several key aspects: the growth trajectory of TinyML publications, identification of leading authors and institutions, examination of co-occurring keywords, and analysis of citation patterns [7]. Through this comprehensive approach, the study aims to reveal the current research trends and future application opportunities of TinyML.

We employed advanced bibliometric tools Biblioshiny (part of the bibliometrix R package) and VOSviewer. Biblioshiny provides a user-friendly web interface for conducting comprehensive bibliometric analyses, while VOSviewer is adept at constructing and visualizing bibliometric networks [8].

The study objectives are to chart the growth of TinyML literature; identify key contributors, including influential authors, institutions, and countries; examine co-occurring keywords trends to explore emerging research trends; and analyze citation patterns that shaped the TinyML

* Corresponding author.
E-mail address: hui.han@ltu.se (H. Han).

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landscape. This study fills a critical gap in the existing literature by providing a comprehensive bibliometric analysis of TinyML, serving as a foundational resource for researchers, practitioners, and policymakers of information systems. Significance lies in its ability to map the current TinyML research, collaborations, identify gaps, and suggest future research and application directions, ensuring the continued growth and impact of this transformative technology.

The paper is organized as follows: Section 2 provides a background on TinyML. Section 3 outlines the research methodology used in the study and presents the initial statistical findings. Section 4 delivers the results of bibliometric analysis. Section 5 presents discussions on the future directions of TinyML research and applications. Finally, Section 6 concludes the paper with an overall summary of the study results, limitations, and future research needs.

2. Background

TinyML integrates machine learning with embedded systems to enable efficient, on-device processing on low-power, resource-constrained devices like microcontrollers and sensors. This technology is increasingly essential in applications requiring real-time data processing, low latency, enhanced privacy, and energy efficiency, as it reduces the dependency on cloud-based solutions. TinyML has demonstrated significant value for information systems across fields such as healthcare, environmental monitoring, industrial automation, and smart cities, where quick, localized data processing is critical for performance, data security, and user privacy [1,4,5,9].

2.1. Low power and cost efficiency

TinyML enables ultra-low-power ML inference on battery-powered devices, significantly reducing both energy consumption and operational costs [4,5,10,11]. Power efficiency is enhanced through various approaches such as strategies such as designing specialized ultra-low-power hardware architectures, leveraging neural co-processors (NCPs) integrated with microcontrollers (MCUs) to maintain on-chip storage of features and weights, and implementing application-specific instruction sets that support agile development and rapid deployment [11,12]. For example, TinyissimoYOLO, a highly flexible and memory-efficient object detection network, enables real-time object detection on microcontrollers with minimal energy usage, achieving high frame rates and using quantized network architectures to maximize efficiency [13]. These innovations in TinyML not only support accurate AI tasks on resource-limited subsystems but also drastically cut power usage, making ultra-low power applications such as object detection and recognition feasible on small, battery-operated devices.

2.2. Enhanced privacy and security

By processing data locally on edge devices, TinyML enhances data security and user privacy, reducing the need for data transmission to cloud services [1,4,10]. Techniques like XimSwap provide real-time video anonymization directly on edge devices, ensuring privacy by design and minimizing computational resources demands [14]. Furthermore, TinyML solutions for presence detection, such as Ultra-wideBand radar-based systems, enhance user privacy by avoiding the capture of sensitive data like images or videos, making them ideal for privacy-sensitive applications [15]. Additionally, TinyFedTL enables federated learning on devices with limited memory, ensuring effective data analysis while preserving user privacy and security [16].

2.3. Real-time processing

TinyML systems enable real-time data processing and decision-making, which is critical for applications like object detection and recognition [10,11]. TinyML boosts real-time processing capabilities by deploying optimized convolutional neural networks (CNNs), such as Tiny-BNN, along with hardware accelerators to perform tasks like image classification in real-time [11,17]. TinyML also utilizes multiple microcontroller platforms for efficient model execution, reducing total inference time by 34.8 % compared to single-controller setups [18]. Additionally, TinyML systems integrate neural co-processors (NCP) to store features and weights on-chip, eliminating latency associated with off-chip memory access [11]. These advancements underscore the feasibility of real-time processing on edge devices, paving the way for innovative applications in IoT, robotics, and augmented reality. By processing data on-device, TinyML supports real-time decision-making without relying on cloud-based computation [19].

2.4. Application in reality

TinyML has emerged as a transformative technology, making it possible to deploy efficient machine learning models on resource-limited devices. This innovation achieves real-time data analytics and edge-based decision-making across multiple fields. In healthcare, TinyML powers continuous and energy-efficient health monitoring by analyzing physiological metrics such as heart rate variability (HRV), to valuable actionable insights into stress levels and sleep quality [20]. Furthermore, it demonstrates significant promise in diagnostic efficacy in pulmonology, with recent studies reporting 96 % accuracy in automatically classifying respiratory diseases through lung sound analysis [21]. For environmental science, TinyML's low energy consumption makes it ideal for persistent ecological monitoring, particularly for air quality assessment, where real-time data acquisition supports data-driven conservation strategies. Within industrial automation, TinyML leverages its ultra-low latency and secure edge computing capabilities to process localized data, ensuring responsive and optimized automation. Regarding smart city, the technology fosters scalable and sustainable urban development through decentralized computational architectures, minimizing reliance on cloud-based systems [22].

3. Research methodology

Bibliometric analysis has emerged as a vital tool in scientific research, playing a pivotal role in evaluating research outputs and identifying trends, particularly in fields like TinyML. This methodology is widely employed to access diverse scientific research dimensions and to rank academic institutions and universities globally [23]. This study applies a bibliometric analysis to explore established and emerging research themes in TinyML. The analysis was conducted using VOSviewer software and the Biblioshiny (a web interface for Bibliometrix), leveraging the Bibliometrix version 3.1 in the R for data processing and visualization [8]. The data for this study was sourced from the WoS database, renowned for its extensive academic coverage. The dataset was saved as a text file to streamline the extraction and analysis of bibliographic information critical for this study. The methodology flow is shown below in Fig. 1.

3.1. Data collection

Articles were collected from the WoS database, which is notable for its comprehensive citation records, encompassing over 34,000 journals and 196 million papers across 254 subject areas [23]. Its compatibility with most bibliometric tools (e.g., Biblioshiny, VOSviewer, CiteSpace) also ensures data integrity.

To compile a relevant dataset for analysis, we employed the search string (tinyml (Topic) OR "tiny machine learning" (Topic)). This search strategy was deliberately designed based on preliminary scoping of the field and expert consultation to capture variations in terminology. Using both the acronym (“TinyML”) and full term (“tiny machine learning”) minimized retrieval bias by ensuring inclusion of studies that may use

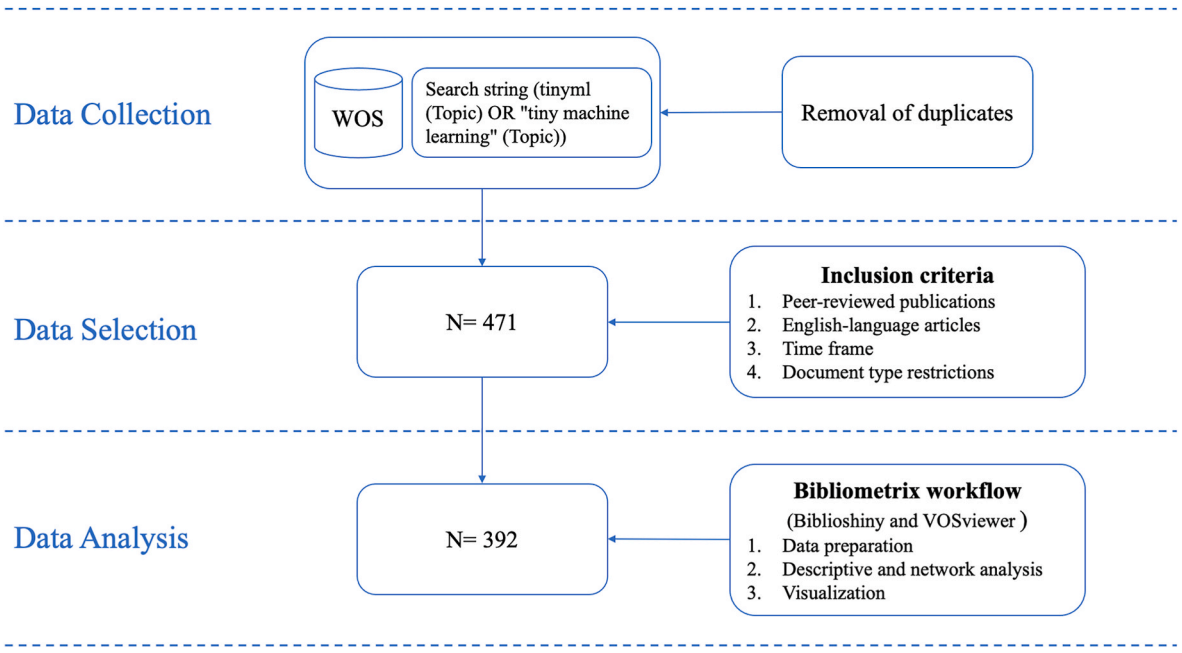


Fig. 1. Methodology flow Diagram.

either form interchangeably. The "Topic" field search—covering the title, abstract, and keywords—was selected to capture studies where TinyML was a research focus rather than a peripheral mention. After removing duplicates, a total of 471 articles were initially identified for further screening.

3.2. Data selection

A structured selection process was conducted to ensure the inclusion of high-quality and relevant research. The following inclusion criteria were applied based on methodological rationale.

1. Peer-reviewed publications: journal articles, conference proceedings, review papers, and early access documents under peer-review were retained. Peer review serves as a fundamental quality control mechanism, ensuring methodological rigor and credibility of reported findings. Non-peer-reviewed materials (e.g., editorials, letters, data papers) were excluded.
2. English-language articles: English remains the predominant language of scientific communication and indexing within WoS. We include English-language publications to ensure accurate interpretation of bibliometric indicators such as keywords, abstracts, and thematic clusters. Non-English papers were excluded because language barriers could lead to misclassification during text-based analyses, potentially biasing results.
3. Time frame (January 1, 2020–May 31, 2024): The selected period reflects the emergence and the rapid development of TinyML as a distinct research area. The term "TinyML" began gaining significant traction since 2020, particularly following the growing convergence of embedded AI. Limiting the time frame to this period allows for an accurate depiction of current research trends.
4. Document type restrictions: The inclusion of journal articles, review papers, and conference proceedings ensured a balanced representation of the field. Review articles were intentionally included because they synthesize existing knowledge, identify thematic patterns, and outline future research directions. Conference proceedings were retained due to their importance in computer science and engineering disciplines, where cutting-edge developments are often first presented at conferences.

After applying these criteria, a total of 392 articles were obtained for analysis. The dataset, downloaded as a text file, included bibliographic details such as author, title, source, abstract, keywords, affiliations, cited references, and funding information.

3.3. Data analysis

Various software tools—such as Bibexcel, CiteSpace, Histcite, VOSviewer, and Gephi–are commonly used in bibliometric studies. The analytical phase employed two complementary bibliometric tools: Biblioshiny (part of the bibliometrix package version 3.1 in R) and VOSviewer. Biblioshiny provides descriptive statistics, trend analysis, and network construction within an open-source environment, while VOSviewer excels in high-quality visualizations and cluster mapping.

The bibliometrix workflow involves three main steps.

1. Data preparation: WoS data were cleaned, standardized, and converted into an R-compatible data format to preserve metadata accuracy.
2. Descriptive and network analysis: Biblioshiny was used for performance analysis (publication trends, citation impact, authorship patterns) and science mapping (co-authorship, co-citation, and keyword co-occurrence networks).
3. Visualization: VOSviewer was employed to visualize scientific knowledge maps, illustrating thematic clusters, collaborative networks, and intellectual structures within the TinyML research area [8].

Reliability was ensured through several measures: First, using a single, high-quality source (WoS) avoided duplication errors and inconsistent citation formats that often arise from cross-database merging. Second, all search strings, inclusion criteria, and data cleaning steps were documented in detail to ensure reproducibility. Third, results obtained from Biblioshiny were cross-validated using VOSviewer to confirm consistency in network structures and thematic clustering. Additionally, random samples of the dataset were manually reviewed to confirm the accuracy of metadata extraction and the relevance of included documents.

4. Results

Bibliometric analysis offers a comprehensive approach to evaluating the growth, impact, and collaboration patterns in scientific research. This section presents descriptive bibliometric findings, authorship trends, influential keywords, citation patterns, and intellectual structures within TinyML research from 2020 to 2024, leveraging Biblioshiny and VOSviewer.

4.1. Overview of TinyML research

TinyML has experienced rapid growth as an emerging interdisciplinary field since its inception around 2019. The documents' average age of 1.52 demonstrated that TinyML is a new field of machine learning. Between 2020 and 2024, the field demonstrated exponential annual growth, averaging 59.23 % increase per year in publications. Out of 392 publications, 240 papers were from distinct journals and conference proceedings, underscoring the growing diversity of dissemination channels within TinyML. On average, each publication has 4.617 citations, indicating the impact and relevance of TinyML research. A total of 1248 authors have contributed to this body of work, with only 11 authors publishing single-authored papers, while the majority (1237 authors) engage in collaborative research. Collaboration is a hallmark of TinyML research, with an average of 4.36 authors per article. Notably, 32.91 % of these collaborations involve international co-authorships, highlighting TinyML's global research networks. Regarding publication types, conference proceedings papers constitute the largest share with 190 documents, followed closely by journal articles at 186, illustrating the broad range of dissemination channels within the TinyML research community. The balance highlights the field's dynamic mix of applied and theoretical research, as well as the newness of the topic. Table 1 summarizes the TinyML research landscape.

Table 1 provides overall insights into the structural characteristics of the TinyML literature. The dataset comprises 12,614 references, indicating that studies in this domain are supported by extensive knowledge of prior research. The large number of author-defined keywords (1,145) with the inclusion of 259 Keywords Plus points to an emphasis on a wide thematic range, spanning both standardized indexing terms and emerging topics. This diversity of keywords reflects an evolving conceptual framework where researchers are continuously introducing new perspectives. Overall, these indicators suggest that TinyML research integrates multidisciplinary nature of the fields (application areas, technologies and methodological approaches), with well-connected intellectual structure supported by an evolving knowledge.

Table 1
Bibliometric analysis summary.

Description	Results
Main information about data	
Timespan	2020–2024
Sources (Journals, Books, etc.)	240
Documents	392
Annual growth rate %	59.23
Document average age	1.52
Average citations per document	4.617
References	12,614
Document contents	
Keywords plus (ID)	259
Authors' keywords (DE)	1145
Authors	
Number of authors	1248
Authors of single-authored documents	11
Authors collaboration	
Single-authored documents	12
Co-authors per document	4.36
International co-authorships %	32.91
Document types	
Article	186
Data paper	1
Early access	4
Proceedings paper	190
Review	11

4.1.1. Growth of publications and citations

The temporal analysis of publications (Fig. 2) revealed a steady increase in both publication numbers and citations. In 2022, the field reached a significant milestone with 194 articles, demonstrating its rising academic and industrial interest [24]. To assess the immediate impact of recent publications, the total citation per year metric is employed, reflecting the relevance and influence of TinyML studies [8]. Citation trends corroborate this growth, showing a parallel increase in the immediate recognition and impact of TinyML research.

4.1.2. Domain categories

The categorization of articles within the WoS database (Fig. 3) highlights the interdisciplinary nature of TinyML research.

- Electrical and Electronics Engineering (228 papers) dominates the field, reflecting investments in hardware optimization and energy-efficient designs.
- Computer Science and Information Systems (100 papers) emphasizes system integration (hardware, software, and operational processes) and real-time processing and automation, often with minimal computational resources.
- Artificial Intelligence (81 papers) showcases the importance of advanced algorithms and neural network optimization.

Other key areas include Computer Science Theory and Methods (76 papers), Telecommunications (74 papers), Instruments and Instrumentation (63 papers), Hardware Architecture (57 papers), Computer Science Interdisciplinary Applications (38 papers), and Computer Science Software Engineering (34 papers). This distribution showcases the interdisciplinary nature of TinyML research and its broad relevance across multiple technological and scientific fields.

4.2. Author productivity through Lotka's law

A deeper analysis of authorship revealed the application of Lotka's Law, which predicts that most authors contributed to a single publication [25]. The core premise of Lotka's law is that higher publication frequency reflects a sustained engagement from authors in each research area [8].

Table 2 and Fig. 4 reveal that 79.5 % of authors have published only a single article, marking the majority as occasional contributors. Conversely, 20.5 % have published two or more articles, with 6.4 % publishing three or more, underscoring a limited but emerging cohort of dedicated contributors in the TinyML domain. This distribution, aligning with Lotka's law, suggests that TinyML is still in its formative stages, as many researchers are initially exploring the field rather than committing to long-term, in-depth research collaborations.

The adherence of TinyML publication patterns to Lotka's law, as shown by the dashed line in Fig. 4, indicates a conventional productivity distribution: a large number of authors with single contributions and a smaller group with multiple donations. This pattern reflects TinyML's status as an emerging field, with researchers currently testing the waters rather than engaging in sustained investigations.

4.3. Author's keywords

The keyword co-occurrence network (Fig. 5) provides insights into the evolving research themes in TinyML. The analysis utilized

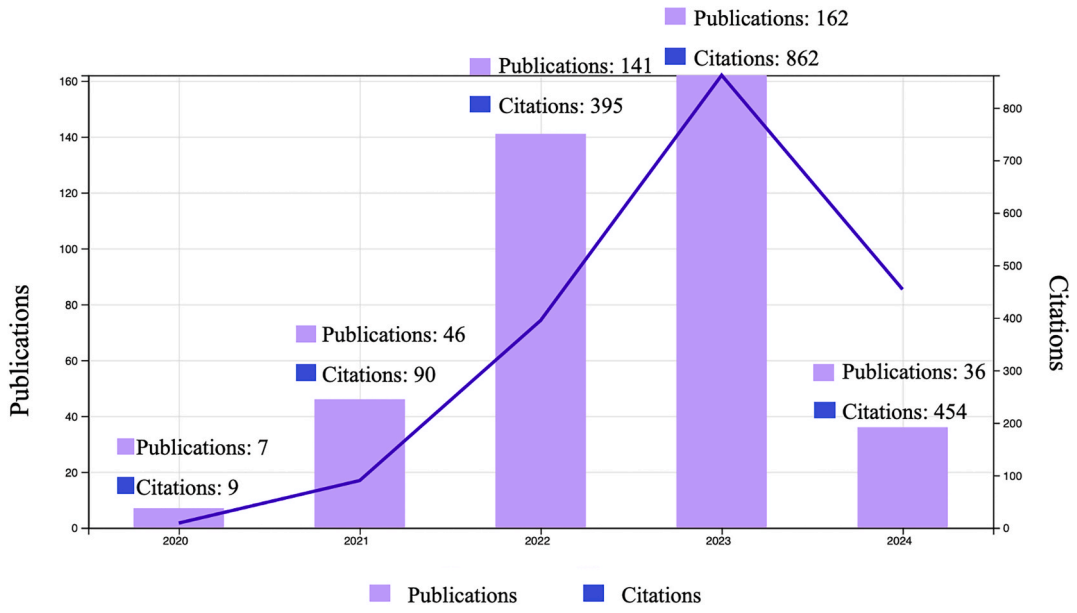


Fig. 2. Year-wise publication and citation trend.

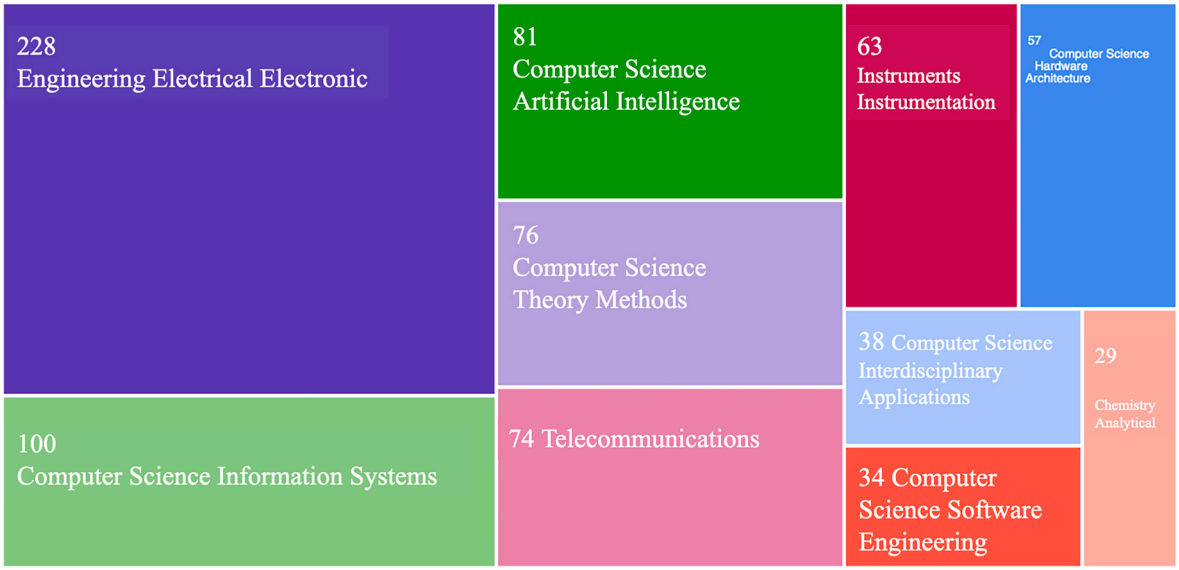


Fig. 3. TreeMap chart of WoS Categories.

Table 2
Author productivity through Lotka's law.

Documents written	N. of Authors	Proportion of Authors
1	992	0.795
2	176	0.141
3	40	0.032
4	17	0.014
5	12	0.010
6	1	0.001
7	2	0.002
8	3	0.002
9	2	0.002

VOSviewer's thesaurus function to standardize terms by merging abbreviations with their full terms. Setting a minimum threshold of three occurrences per keyword, we identified 106 keywords from an initial 1000 for visualization items in the network analysis.

In this network, circle size represents the frequency of keyword occurrences, with larger circles indicating more frequent keywords (Fig. 5) and therefore the evolving themes in TinyML.

- **Core Keywords**: terms like "tiny machine learning" (occurring 286 times, total link strength (TLS) of 740), "machine learning" (77 occurrences and a TLS of 298), and "Internet of Things" (76 occurrences and a TLS of 179) dominate the network.
- **Emerging Themes**: "microcontroller" (51 occurrences and a TLS of 176), and "edge computing" (51 occurrences and a TLS of 176), are relatively prominent, illustrating the field's focus on integrating AI into constrained environments.

These key terms and their connections underscore the core themes within TinyML research, reflecting a strong focus on machine learning, IoT, microcontrollers, and edge computing. The interconnectedness of these topics suggests a robust interest in advancing applications that

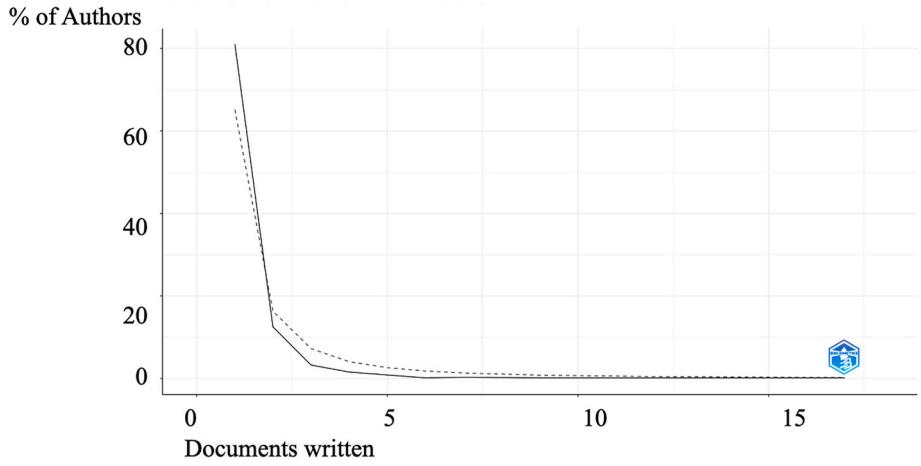


Fig. 4. Author productivity analysis by Lotka's law.

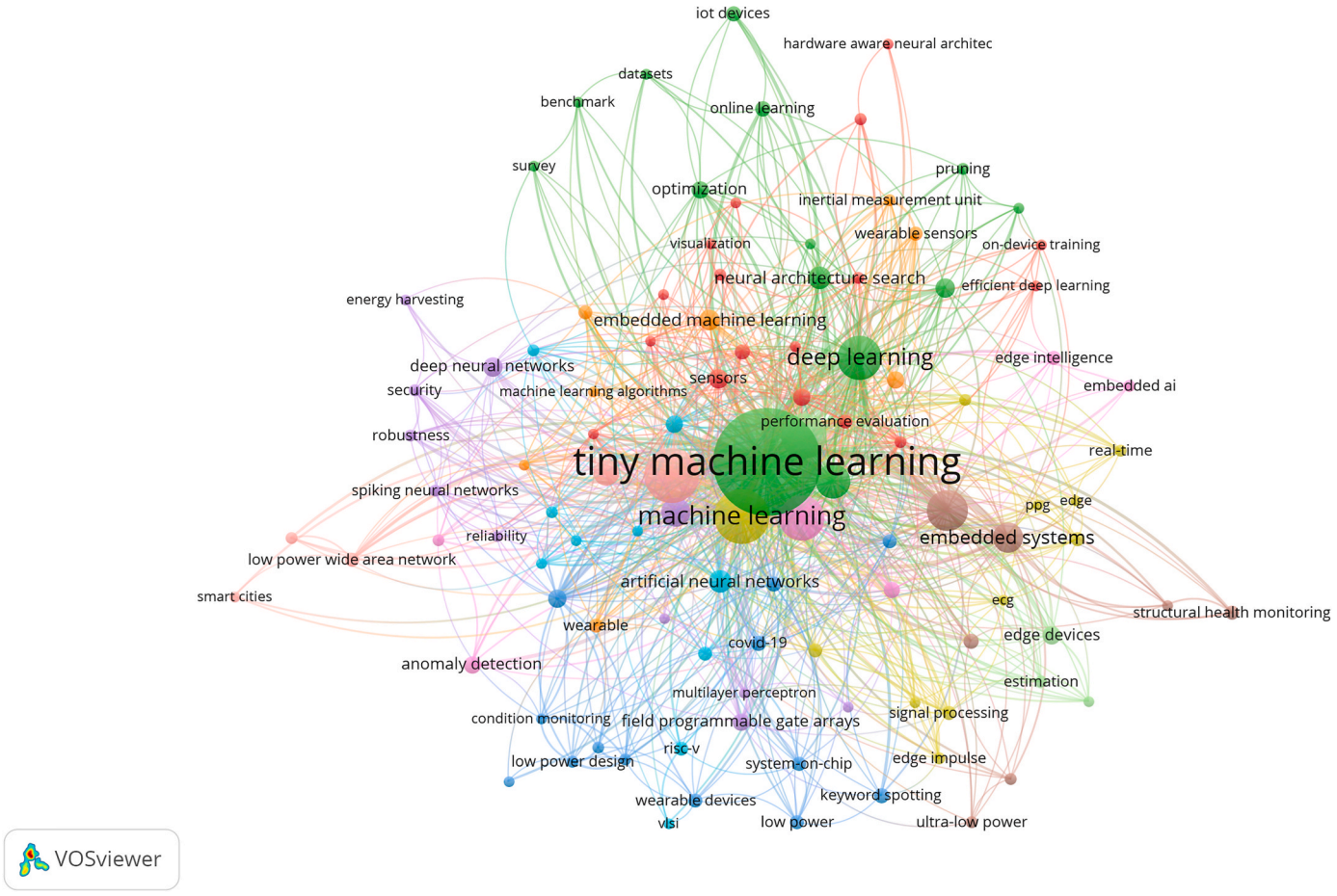


Fig. 5. Authors' keywords Co-occurrence.

combine ML capabilities with resource-constrained devices in decentralized environments.

4.4. Citation analysis by countries

The country-wise citation structure data for TinyML research, sourced from the WoS database, indicates that 60 countries are involved in this field. Applying a threshold of at least one publication and one citation using VOSviewer [26], 56 countries met the inclusion criteria.

In terms of total citations, Italy leads with 397, followed by India with 330, Switzerland with 282, and USA with 227 (see Fig. 6 and Table 3). Notably, when normalized number by publication volume, India ranks highest, underscoring a substantial research impact relative to its number of publications.

Table 3 also highlights that, countries with high publication counts, such as Italy, the USA, Switzerland, and India, generally have the highest citation totals [26]. However, exceptions exist; for instance, England, despite its large publication volume, received fewer citations than Germany and Spain, which have fewer publications but higher citation rates.

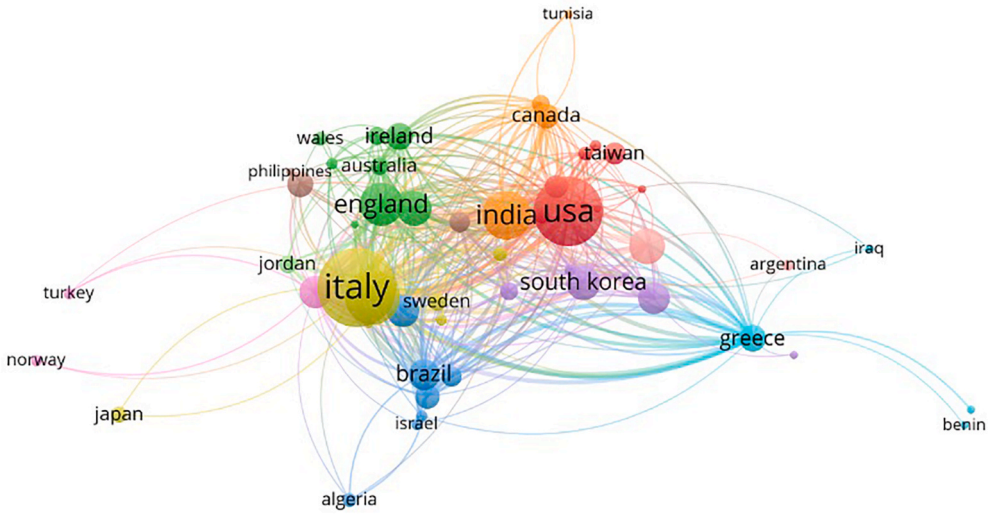


Fig. 6. Total citation analysis for countries.

Table 3
Top 15 contributing countries.

Countries	Citations	Documents	Norm. citations	Avg. citations
Italy	397	92	60.1317	4.3152
India	330	35	85.2095	9.4286
Switzerland	282	44	26.7179	6.4091
USA	227	70	62.5759	3.2429
Spain	177	16	17.8757	11.0625
Germany	121	16	14.0592	7.5625
England	103	27	31.9027	3.8148
Greece	86	11	15.6029	7.8182
Brazil	65	14	9.1902	4.6429
Egypt	60	5	25.2737	12
United Arab Emirates	59	8	6.5373	7.375
Nigeria	57	2	5.0123	28.5
Saudi Arabia	53	15	18.8485	3.5333
China	47	19	18.479	2.4737
France	40	17	10.0045	2.3529

An analysis of average citations per document for the top 15 countries reveals notable variability. Nigeria has the highest average, with 28.5 citations per document, whereas France has the lowest, averaging 2.35 citations per document. This variation illustrates the differing research impact among countries within the TinyML domain, reflecting both publication volume and citation quality.

4.5. Intellectual structure: historiography

A chronological historiographic analysis was conducted using Biblioshiny to access and map the intellectual evolution of TinyML research, identifying pivotal articles and their influence in the field [27]. The analysis identified the 20 most influential publications, with their specific contributions summarized in Table 4 and their interconnected relationships visualized in Fig. 7. The historiographic map in Fig. 7 illustrates how foundational works have influenced more recent research, with nodes representing individual articles arranged by publication year. Dashed arrows between nodes signify citation links, typically directed toward earlier works, while the circle sizes denote the Local Citation Score (LCS) and Global Citation Score (GCS), indicating the articles' relative influence.

Among these, the paper "*TinyML-enabled Frugal Smart Objects: Challenges and Opportunities*" by Sanchez-Iborra R (2020) stands out as the most influential, with 134 GCS and 76 LCS, establishing its foundational role in TinyML research. Similarly, Wang XY's "*FANN-on-MCU: An Open-Source Toolkit for Energy-Efficient Neural Network Inference at the Edge of the Internet of Things*" (2020) ranks second, with 94 GCS and 25 LCS, emphasizing practical advancements in resource-efficient neural network deployment. The third-ranked article, "*A Review on TinyML: State-of-the-Art and Prospects*" garnered 93 GCS and 53 LCS, providing a comprehensive overview of the field's status and future potential.

These 20 influential articles span diverse applications, including embedded devices [28–30], IoT [19,31,32], autonomous vehicles [33], agriculture [34], environmental monitoring [35], and healthcare [36]. This historiographic analysis underscores the interdisciplinary nature and growing impact of TinyML research, positioning it as a critical enabler for innovative solutions across sectors.

5. Future direction for TinyML research and applications

The future of TinyML research appears remarkably promising, presenting numerous opportunities for transformative innovations and diverse applications. As the field rapidly evolves, it becomes crucial to identify key research trajectories to overcome existing challenges and effectively harness emerging opportunities. To systematically summarize these trends, we synthesize insights from eleven literature review and survey papers. These review papers were chosen to ensure that the research directions are grounded in comprehensive perspectives. Based on this analysis, we distil nine key future research directions that capture the most prominent thematic areas and reflect the multifaceted nature of TinyML research. Table 5 presents an overview of the primary research themes identified across the eleven literature review papers, while Table A1 in the Appendix provides a detailed topic information.

5.1. Advanced technologies integration

TinyML's potential for innovation hinges on its ability to integrate seamlessly with emerging technologies. These integrations will enhance its applications in domains ranging from healthcare to industrial automation.

5.1.1. Advanced sensor integration

Future advancements in TinyML should leverage multimodal sensing platforms that incorporate advanced sensor technologies such as the Inertial Measurement Unit (IMU), audio, image, and barometer sensors (temperature, humidity, and pressure). These sensors enable comprehensive data collection, improving real-time decision-making in context-aware applications with higher precision applications like

Table 4
The Top 20 articles with their LCS and GCS.

Paper	Title	GCS	LCS
Sanchez-Iborra R, 2020	Tinyml-enabled frugal smart objects: challenges and opportunities	134	76
Wang XY, 2020	Fann-on-mcu: an open-source toolkit for energy-efficient neural network inference at the edge of the internet of things	94	25
Ray PP, 2022	A review on tinyml: state-of-the-art and prospects	93	53
Dutta L, 2021	Tinyml meets iot: a comprehensive survey	74	51
Capotondi A, 2020	Cmix-nn: mixed low-precision cnn library for memory-constrained edge devices	64	16
Ajani TS, 2021	An overview of machine learning within embedded and mobile devices-optimizations and applications	57	7
Shafique M, 2021	TinyML: current progress, research challenges, and future roadmap	36	18
Signoretti G, 2021	An evolving tinyml compression algorithm for iot environments based on data eccentricity	28	14
Schizas N, 2022	TinyML for ultra-low power AI and large scale IoT deployments: A systematic review	25	10
Ren Hy, 2021	Tinyol: tinyml with online-learning on microcontrollers	22	17
De Prado M, 2021	Robustifying the deployment of tinyml models for autonomous mini-vehicles	22	16
Alajlan NN, 2022	Tinyml: enabling of inference deep learning models on ultra-low-power IoT edge devices for ai applications	21	13
Zaidi SAR, 2022	Unlocking edge intelligence through tiny machine learning (TinyML)	21	8
Tsoukas V, 2021	A review of machine learning and tinyml in healthcare	19	15
Ravaglia L, 2021	A tinyml platform for on-device continual learning with quantized latent replays	19	10
Alongi F, 2020	Tiny neural networks for environmental predictions: an integrated approach with Miosix	18	13
Bian SZ	Capacitive sensing based on-board hand gesture recognition with tinyml	17	10
Andrade P, 2022	A tinyml soft-sensor approach for low-cost detection and monitoring of vehicular emissions	16	11
Abadade Y, 2023	A comprehensive survey on TinyML	14	8
Sanchez-Iborra R, 2021	LPWAN and embedded machine learning as enablers for the next generation of wearable devices	13	7
Nicolas C, 2022	TinyML smart sensor for energy saving in internet of things precision agriculture platform	12	8
Raza W, 2021	Energy-efficient inference on the edge exploiting TinyML capabilities for UAVs	12	7
Doyu H, 2021	A TinyMLaaS ecosystem for machine learning in IoT: overview and research challenges	8	7
Sudharsan B, 2021	An SRAM optimized approach for constant memory consumption and ultra-fast execution of ML classifiers on TinyML hardware	7	7

environmental monitoring and healthcare [2]. For instance, barometric sensors can enhance trend predictions in climate studies, while integrated healthcare sensors provide a holistic view of patient health real-time vitals monitoring [41]. Developing systems that effectively combine such sensors will unlock new capabilities in resource-constrained environments.

5.1.2. *Edge-to-cloud continuum*

The edge-to-cloud continuum is a critical research direction for TinyML, enabling dynamic workload distribution. Computationally intensive tasks such as model training can occur in the cloud, while real-time inferences are executed on edge devices [41]. This reduces latency and optimizes resource usage, benefiting applications like industrial IoT (IIoT), where real-time machinery monitoring and predictive maintenance can minimize downtime. Similarly, smart city initiatives can leverage this model for efficient traffic management, energy use, and public safety [3]. Research into seamless coordination between edge and cloud systems will maximize TinyML's performance in these areas.

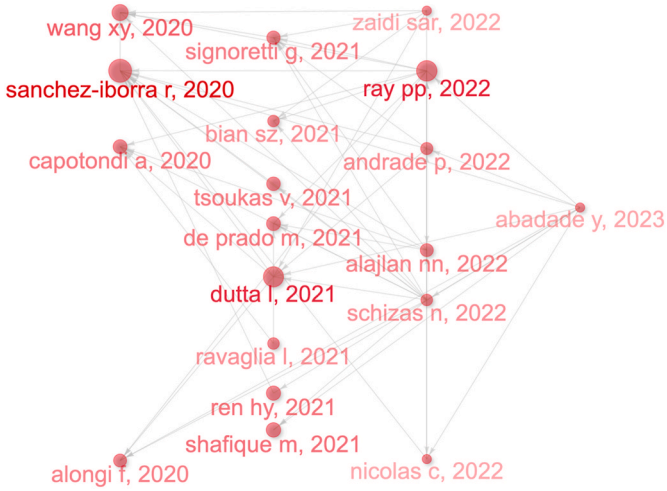


Fig. 7. Historiography distribution in TinyML.

5.1.3. *Augmented reality (AR) and virtual reality (VR) integration*

Integrating TinyML into AR and VR systems can redefine user experiences. TinyML's localized processing supports real-time object recognition and interaction, which is essential for applications like surgical training and rehabilitation. Lightweight, energy-efficient AR/VR devices powered by TinyML will enable wider adoption in education and entertainment. Future studies should optimize these systems to balance real-time performance and resource constraints, ensuring scalability and efficiency.

5.2. *Sustainable TinyML*

With increasing awareness of AI's environmental impact, TinyML has a role in driving sustainable AI innovation.

5.2.1. *Energy-efficient algorithms*

Techniques like pruning, quantization, and knowledge distillation can reduce energy consumption in TinyML models [28]. Spiking neural networks, mimicking biological neurons, present an energy-efficient alternative for real-time tasks. Research should focus on balancing computational efficiency with energy constraints to scale TinyML for diverse applications.

5.2.2. *Eco-friendly hardware*

Developing recyclable and biodegradable microcontrollers and sensors is essential for sustainable TinyML. Innovations in material science and clear recycling standards for TinyML devices will help address e-waste challenges. Collaboration between academia and industry is critical to advancing these sustainable hardware solutions.

5.2.3. *Energy harvesting*

TinyML systems can harness ambient energy sources, such as solar or kinetic energy, to operate without traditional power supplies [40]. This is particularly valuable for remote deployments in environmental monitoring or wearable healthcare devices. Research into energy-harvesting methods will enhance TinyML's feasibility in resource-limited settings.

5.3. *Human-centric TinyML*

Future TinyML systems should prioritize human-centric designs, personalization, and custom hardware tailored to specific use cases [38].

User-centric design: Smart homes powered by TinyML could autonomously adjust lighting, temperature, and energy consumption based on user preferences. Future research in this area should focus on enhancing

Table 5
Topics summary from review literature.

Topics	Papers										
	[2]	[6]	[4]	[37]	[38]	[39]	[40]	[41]	[28]	[3]	[42]
IoT	✓				✓	✓	✓			✓	
Benchmarking		✓	✓		✓		✓				✓
Memory		✓		✓							
On-device				✓			✓			✓	
Heterogeneity		✓		✓				✓			✓
Privacy and security		✓		✓			✓	✓	✓	✓	
Frameworks	✓		✓		✓						
Optimization									✓		✓
Ethical AI			✓								
Energy efficiency					✓		✓		✓		✓
Tech integration			✓					✓			✓
Applications		✓	✓		✓	✓	✓		✓		✓
Models	✓	✓		✓	✓			✓	✓		
Influencing factors			✓				✓				
Offloading		✓	✓		✓					✓	
ML algorithm						✓	✓			✓	✓
In-memory											
Embedded systems	✓		✓	✓	✓				✓		✓
Wearable devices				✓							✓
Mobiles			✓						✓		
Microcontrollers	✓							✓			
Real-time					✓						✓

contextual awareness, allowing systems to continuously learn from user behavior, environmental cues, and temporal patterns while operating under strict resource constraints. Intelligent adaptation mechanisms, such as on-device personalization, incremental learning, and adaptive control strategies, can enable smart home systems to evolve with changing user needs without frequent manual configuration. Addressing challenges related to model accuracy, energy efficiency, and robustness will be essential to delivering seamless and trustworthy user experiences.

Custom hardware, such as application-specific integrated circuits (ASICs) and neuromorphic chips, can significantly improve the efficiency and performance of TinyML devices [43]. These innovations will enable faster, more energy-efficient processing for healthcare and environmental monitoring.

Personalization is a key trend, enabling TinyML systems to deliver tailored healthcare, assistive technologies, and adaptive smart environments. Using user-specific data, TinyML delivers customized services, such as personalized health monitoring. Wearable health devices with TinyML capabilities could track key health metrics continuously, providing insights customized to each user's needs [42]. TinyML also has transformative potential for accessibility through assistive technologies like speech recognition, gesture detection, and visual recognition on small IoT devices. Future studies should explore methods, such as adaptive algorithms that adjust to individual health patterns, to develop personalized TinyML models that cater to diverse user requirements.

5.4. Ethical and regulatory landscape

The advancement of TinyML requires robust ethical and regulatory measures to ensure responsible innovation [4]

Bias mitigation: Bias in TinyML could lead to disparities in healthcare, finance, or public safety applications [44]. Research into unbiased data collection and transparent algorithms will enhance model fairness and user trust.

Societal implications: Understanding TinyML's implications for employment, privacy, security, and equity will guide its ethical deployment [41]. Policymakers, technologists, and community leaders must collaborate to align TinyML innovations with societal needs.

5.5. Collaboratory efforts

Interdisciplinary collaborations are essential to address TinyML's complex challenges and expand its applications.

Cross-disciplinary research: Engaging experts from diverse fields such as healthcare, environmental science, agriculture, computer science, electrical engineering, data science, and specialized areas can drive innovative solutions for specific industry challenges, such as precision farming or real-time diagnostics [38].

Industry-academia partnerships: Partnerships between academia and industry can accelerate the transition from research to real-world solutions [27], ensuring that TinyML technologies are practical and scalable. Such collaborations are particularly important for TinyML, where successful deployment depends not only on model accuracy but also on hardware compatibility, energy efficiency, and long-term maintainability. Joint research initiatives, co-developed prototypes, and shared testbeds can facilitate iterative refinement of TinyML solutions and ensure their robustness in operational environments. Furthermore, industry–academia partnerships can support standardisation efforts, workforce training, and technology transfer, ultimately enabling scalable and sustainable adoption of TinyML across sectors such as smart manufacturing, healthcare, and IoT ecosystems.

Open-source ecosystem: Open-source TinyML frameworks [2] and repositories encourage collaboration, enabling researchers and developers to build on shared innovations. Competitions and hackathons can further advance the state of the art in TinyML.

5.6. Transformative applications

TinyML offers transformative potential across emerging markets, addressing resource limitations through specialized solutions that promote sustainable development.

5.6.1. Smart cities

As urbanization accelerates, TinyML has the potential to revolutionize smart city infrastructures by optimizing resource management in real time. Through integration with urban sensors, TinyML can improve traffic flow, reduce waste, and enhance energy efficiency, creating cities that are not only smarter but also more sustainable [45]. Future research should prioritize the scalability and real-time processing capabilities of TinyML for large-scale applications. This is the development of

algorithms that manage extensive datasets efficiently while seamlessly integrating with legacy urban systems to ensure a smooth transition to smarter city environments [42].

5.6.2. Environmental monitoring

TinyML-enabled sensors are essential for continuous environmental monitoring, especially in remote areas or inaccessible regions. These advanced sensors can track critical metrics such as air and water quality, wildlife behavior, and ecological changes in real time, enabling proactive conservation efforts and minimizing the human impact on natural ecosystems [46]. The integration of TinyML with satellite and drone technologies can further enhance the scope of ecosystem management, enabling real-time, large-scale interventions. To meet global environmental challenges, TinyML solutions must remain cost-effective while being adaptable to diverse ecological scenarios, thereby fostering a sustainable future.

5.6.3. Smart agriculture

TinyML is poised to transform agriculture through precision farming techniques and enhanced pest management. By employing TinyML sensors to monitor real-time data such as soil moisture, nutrient content, and weather conditions, farmers can make data-driven decisions that optimize irrigation, fertilization and overall crop management [47]. Future research should focus on integrating TinyML with cutting-edge agricultural technologies to promote sustainable practices and maximize yield. Collaboration among stakeholders, agricultural experts, technology providers, and farmers, is critical to tailoring solutions to specific regional needs and achieving widespread adoption.

5.6.4. Healthcare

In the healthcare sector, TinyML offers transformative possibilities for patient monitoring, diagnostics, and treatment, especially in underserved and remote areas. The development of affordable, portable TinyML diagnostic devices can strengthen public health initiatives by providing access to vital medical resources where they are most needed [36]. Wearable TinyML devices capable of continuously monitoring key health metrics such as heart rate, blood pressure, and glucose levels, can deliver real-time alerts and personalized health recommendations, empowering patients to manage their conditions proactively [48]. Future research should focus on creating robust, secure.

TinyML systems that safeguard patient data while integrating seamlessly with telemedicine platforms to improve remote healthcare accessibility and outcomes globally.

5.7. Resilient systems

Ensuring TinyML systems are resilient to challenges like hardware malfunctions and cyberattacks is critical for long-term reliability.

Self-healing mechanisms: Systems equipped with self-healing capabilities can detect and address issues autonomously, ensuring consistent performance in high-stakes applications such as healthcare and industrial automation [4]. Future research should explore lightweight fault-detection techniques, adaptive recovery strategies, and on-device diagnostics that can operate within the stringent computational and energy constraints of TinyML platforms. Approaches such as continual monitoring, redundancy-aware model execution, and adaptive reconfiguration can enable systems to maintain consistent performance even under changing operating conditions or partial system failures. By enhancing robustness, fault tolerance, and long-term autonomy, self-healing TinyML systems can significantly improve trustworthiness and enable wider adoption in mission-critical real-world deployments.

Adaptive models: Adaptive models represent a critical advancement for enabling robust and flexible TinyML systems in dynamic and unpredictable environments. Models that can evolve over time with minimal human intervention allow TinyML-enabled devices to remain effective despite changes in operating conditions, data distributions, or user requirements. Such adaptability is particularly important in application domains like agriculture, where weather patterns, soil conditions, and crop growth stages vary continuously, and in healthcare, where patient conditions and physiological signals can change rapidly [6].

5.8. Evolutionary frameworks and models

The advancement of TinyML frameworks and models is crucial for enabling machine learning capabilities on resource-limited devices. Future research should prioritize enhancing existing frameworks and creating models specifically tailored to meet the unique demands of TinyML applications.

5.8.1. Evolutionary frameworks

TinyML frameworks must facilitate ML deployment on compact, resource-constrained devices like microcontrollers, addressing critical challenges such as limited power, real-time processing needs, and stringent security requirements [38]. Future work should focus on creating scalable frameworks capable of evolving with microcontroller technology while optimizing energy-efficient for battery-dependent systems. Additionally, ensuring secure data handling and enabling rapid, low-power processing are key to making TinyML's frameworks practical and reliable for diverse real-world applications.

5.8.2. Evolutionary models

Future TinyML models should leverage federated learning, a technique that enables distributed model training across IoT devices while safeguarding data privacy [28]. This approach helps address major challenges, including reducing power consumption, managing limited storage capacity, and compressing models to suit resource-constrained environments. Furthermore, incorporating self-learning capabilities into TinyML models would enhance adaptability, enabling these systems to adjust autonomously to dynamic and evolving conditions [16]. Transfer learning techniques, which allow models to utilize pre-trained knowledge, can significantly minimize the need for extensive datasets and computing power, making TinyML solutions more versatile and applicable across diverse scenarios [49].

5.9. Security and privacy measures enhancements

Robust security and privacy measures are essential for safeguarding TinyML applications, ensuring data protection, model integrity, and user confidentiality within resource-constrained devices. Strengthened security will not only mitigate risk but also foster user trust, promoting widespread adoption of TinyML technologies [50].

5.9.1. Lightweight encryption techniques

Developing lightweight cryptographic algorithms optimized for TinyML's constrained resources is a promising direction for future research. These algorithms must provide robust data integrity, authentication, and confidentiality without overburdening the device's computational capabilities [39]. Cryptographic techniques such as elliptic curve cryptography (ECC) and homomorphic encryption can be fine-tuned for TinyML, delivering secure processing while maintaining efficiency. Additionally, designing custom cryptographic protocols tailored specifically to TinyML's limitations will be critical for ensuring security and data privacy across various applications.

5.9.2. Anomaly detection

Anomaly detection mechanism is a vital application of TinyML, enabling the identification of irregular activities that may indicate security threats or technical malfunctions [38]. This capability is particularly important in sensitive fields such as healthcare, automotive, and industrial IoT, where real-time detection is critical. Future models should integrate advanced unsupervised anomaly detection methods to distinguish effectively between normal and abnormal activities,

addressing emerging cyber threats and operational disruptions [51]. These models must be designed for efficient operation on edge devices, ensuring high accuracy and low latency despite resource limitations.

6. Conclusions

6.1. Summary of bibliometric analysis of TinyML

Bibliometric analysis, conducted on TinyML research from 2020 to 2024 using the WoS database, provides a comprehensive examination of trends, key themes, and influential contributions in this rapidly evolving field. Biblioshiny and VOSviewer facilitated the analysis, offering visualizations of the intellectual and collaborative networks that define the TinyML research landscape [52].

The dataset, comprising 392 articles from 240 distinct journals and conference proceedings, underscores the field's growing prominence, with an average of 4.617 citations per document. Collaboration is extensive, averaging four authors per paper, and 32.91 % of contributions involve international co-authorships. Consistent with Lotka's law, 79.5 % of authors contributed only a single publication, reflecting the field's early stage and a broad influx of initial contributions.

Keyword analysis identified dominant themes, including "machine learning," "internet of things," "microcontroller," and "edge computing", showcasing TinyML's interdisciplinary nature. Citation analysis highlighted Italy, India, Switzerland, and USA as leading contributor countries, with normalized citation counts revealing impactful research from countries with fewer publication volumes, such as India and Nigeria. This suggests that impactful research in TinyML is not necessarily tied to the quantity of publications but rather to the quality and relevance of the work produced. Historiographic mapping identifies critical influential studies, such as Sanchez-Iborra's work on TinyML-enabled smart objects and Wang's FANN-on-MCU toolkit, that shape TinyML's current research direction.

Looking forward, TinyML research should focus on several key areas to support its continued growth and maximize its impact. First, integrating advanced technologies, including sophisticated sensors, the edge-to-cloud continuum, and augmented reality/virtual reality (AR/VR), to broaden the range of applications and bring powerful, context-aware capabilities to TinyML. Second, developing sustainable and environmentally conscious AI practices will minimize the carbon footprint of TinyML, promoting green technology in resource-constrained environments. Third, prioritizing strong security and privacy measures is essential to protect data integrity and foster user trust, particularly as TinyML becomes embedded in sensitive domains like healthcare and finance. Fourth, creating ethical guidelines to ensure transparency, fairness, and accountability will support responsible AI practices in TinyML deployment, addressing potential biases and promoting equitable usage.

In addition, fostering interdisciplinary collaborations will be crucial for driving innovation and meeting the unique challenges of TinyML, as partnerships between computer scientists, engineers, and domain experts can result in groundbreaking applications. Tailoring TinyML solutions to specific fields, such as smart agriculture, healthcare, and smart cities, will help optimize resources and enhance sustainability. Continuous improvements to frameworks and models will be necessary to ensure scalability, interoperability, and the ability to incorporate self-learning algorithms. Developing resilient systems with enhanced security and privacy features is also essential for reliable deployment in real-world environments. Expanding interdisciplinary research, developing adaptive frameworks, and improving model resilience will also be necessary in supporting TinyML deployment in diverse, resource-constrained environments.

In conclusion, this bibliometric analysis provides an in-depth overview of TinyML research. It highlights the rapid growth, collaboration patterns, and key research themes within the field. By identifying influential studies and outlining future directions, this study provides a valuable resource for researchers, practitioners, and policymakers aiming to explore further and contribute to the field of TinyML.

6.2. Limitations of the study

While this study provides a comprehensive bibliometric analysis of TinyML, it has some limitations. First, relying exclusively on the WoS database narrows the scope, potentially omitting relevant publications from other databases. Expanding the sources would enhance the comprehensiveness of the analysis. Second, the five-year time frame (2020–2024) may limit the ability to capture long-term trends and evolution in the field as TinyML technology continues to evolve. Expanding the sources would enhance the comprehensiveness of the analysis. Finally, while this study emphasizes quantitative metrics like citation patterns and author collaborations, it does not delve deeply into the qualitative aspects of research impact or the specific challenges in applying TinyML across diverse domains. Future research should address these gaps by incorporating qualitative analyses, assessing the practical outcomes of TinyML innovations, and exploring broader societal implications. By addressing these limitations, future studies can offer a more holistic understanding of TinyML's development, enabling a nuanced evaluation of its long-term potential across various applications and fields.

CRediT authorship contribution statement

Hui Han: Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Silvana Trimi:** Writing – review & editing, Supervision, Conceptualization. **Sang M. Lee:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.array.2025.100674.

Data availability

No data was used for the research described in the article.

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